Monday Warm Up: Unit 4, Mutual Inductance, Self-inductance, and RL circuits

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1 Memory Bank

- $\epsilon = -L\Delta I/\Delta t$ ... Faraday's Law
- $\epsilon(t) = \epsilon_0 \sin(\omega t)$ ... AC voltage generated by generator. ϵ_0 is proportional to ω.
- $P_{max} = V_{max}^2/R$ ... Max power of an AC generator.
- $P_{ave} = \frac{1}{2}P_{max}$ ... Average power of an AC generator.
- $T = 1/f$ ... Inverse relationship between frequency and period.
- 1 H = 1 Ω s ... The *Henry* is a unit of inductance, often represented by L.
- $I(t) = I_0\left(1 - \exp(-t/\tau)\right)$ ... Current when RL circuits turn on ($\tau = R/L$).
- $I(t) = I_0 \exp(-t/\tau)$ ... Current when RL circuits turn off ($\tau = R/L$).

2 Mutual Inductance

1. Consider Fig. 1. If the coils of the heating element for an electric dryer are *counter-wound*, why does this reduce back emf when the current is increased?

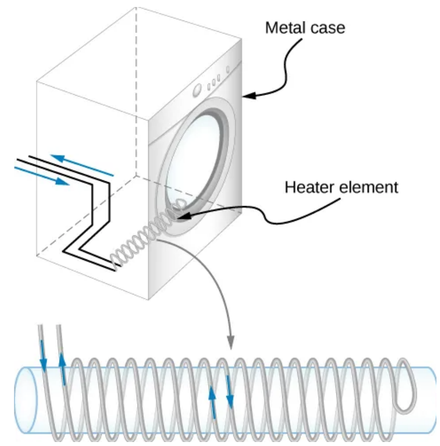

Figure 1: Coils wrapped around the outside of a solenoid.

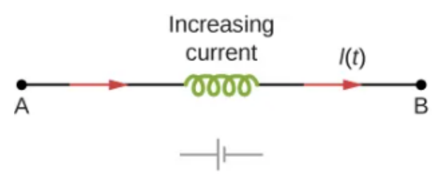

Figure 2: Back emf of an inductor with increasing current.

2. What is the inductance of a device, if we measure a 20 mV back emf for an increasing current of 100 mA per second?

3 Self-Inductance

1. Consider Fig. 2. (a) What is the back emf in across an inductor with $L = 50$ mH, if the current is increased from 0 to 2 A in 0.2 seconds? (b) What is the emf across the inductor if the current is shut off in 0.2 seconds? (d) How do you think it works if two inductors are in series?

4 RL Circuits

1. (a) What is the characteristic time constant for a 9.0 mH inductor in series with a 3.00 Ω resistor? (b) Find the current 6.00 ms after the circuit is closed. (c) What will be the current if the circuit is left closed indefinately?